NOETHERIAN CLOSURE PROPERTIES OF THE CATEGORY C-PRIME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reference environment citing EGA1 Proposition 7.5.5 is statement-side metadata; part (2) assembles Noetherianity of the quotient from the same permanence lemma used in part (4)

Editorial note. The label lemma-topologically-finite-type-Noetherian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Algebra, Lemma algebra-lemma-completion-Noetherian-Noetherian: the I-adic completion of a Noetherian ring is Noetherian
- Algebra, Lemma algebra-lemma-Noetherian-permanence: a polynomial ring over a Noetherian ring is Noetherian
- Lemma lemma-topologically-finite-type (delivered, L48111): every object of C-prime is the I-adic completion of a finite type A-algebra
- Algebra, Lemma algebra-lemma-completion-tensor: for B in C-prime every finite B-module is IB-adically complete

Source reference. Original lemma 00FN: sources/stacks-chow/tags.txt, source lines 577–577. Prerequisite pointer invoked at 416; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0316: sources/stacks-chow/tags.txt, source lines 3973–3973. Prerequisite pointer invoked at 414; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00MA: sources/stacks-chow/tags.txt, source lines 816–816. Prerequisite pointer invoked at 420; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at 402; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0AJP: sources/stacks-restricted/pinned-original-source.tex, source lines 346–346. Prerequisite pointer invoked at 417; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring and let $I \subset A$ be an ideal. Then*

(1) *every object of the category $\mathcal{C}'$ (0AL4) is Noetherian,*
(2) *if $B \in \mathrm{Ob}(\mathcal{C}')$ and $J \subset B$ is an ideal, then B/J is an object of $\mathcal{C}'$,*
(3) *for a finite type A-algebra C the I-adic completion $C^\wedge$ is in $\mathcal{C}'$,*
(4) *in particular the completion $A[x_1, \ldots, x_r]^\wedge$ is in $\mathcal{C}'$.*

Proof. Part (4) follows from Algebra, Lemma 0316 as $A[x_1, \ldots, x_r]$ is Noetherian (Algebra, Lemma 00FN). To see (1) by Lemma 0AJP we reduce to the case of the completion of the polynomial ring which we just proved. Part (2) follows from Algebra, Lemma 00MA which tells us that ever finite B-module is IB-adically complete. Part (3) follows in the same manner as part (4). □